ACTG175_data_summary.csv

group	n_total	n_observed_y	n_missing_y	missing_pct	observed_mean	observed_sd	observed_variance
0	532	321	211	39.662	287.617	166.383	27683.41
1	522	333	189	36.207	341.252	173.585	30131.86
2	524	337	187	35.687	354.819	173.369	30056.80
3	561	351	210	37.433	328.792	178.255	31774.87

ACTG175_dataset_info.csv

dataset	n_total	n_groups	natural_missing_pct	observed_variance_ratio
ACTG175	2139	4	37.26	1.148

ACTG175_missingness_logistic_model.csv

term	Estimate	Std. Error	z value	Pr(>|z|)
(Intercept)	−0.331	0.339	−0.978	0.328
group1	−0.143	0.127	−1.122	0.262
group2	−0.162	0.128	−1.267	0.205
group3	−0.094	0.125	−0.751	0.453
cd40	−0.001	0.000	−2.160	0.031
cd80	0.000	0.000	−0.766	0.444
age	−0.005	0.005	−0.925	0.355
wtkg	0.006	0.003	1.748	0.081

ACTG175_natural_missing_results.csv

scenario	missing_rate	method	contrast	estimate	se	lower	upper	width
Natural missingness (37.3%)	0.373	Complete−case	1 vs 0	53.635	13.294	22.348	84.923	62.575
Natural missingness (37.3%)	0.373	Complete−case	1 vs 0	53.635	13.528	21.859	85.411	63.552
Natural missingness (37.3%)	0.373	Complete−case	2 vs 0	67.202	13.245	36.030	98.375	62.345
Natural missingness (37.3%)	0.373	Complete−case	2 vs 0	67.202	13.489	35.519	98.885	63.366
Natural missingness (37.3%)	0.373	Complete−case	3 vs 0	41.175	13.295	9.884	72.466	62.582
Natural missingness (37.3%)	0.373	Complete−case	3 vs 0	41.175	13.357	9.802	72.549	62.747
Natural missingness (37.3%)	0.373	PMM MI	1 vs 0	62.620	13.054	31.947	93.292	61.345
Natural missingness (37.3%)	0.373	PMM MI	1 vs 0	62.620	13.129	31.800	93.440	61.639
Natural missingness (37.3%)	0.373	PMM MI	2 vs 0	70.268	13.020	39.676	100.861	61.186
Natural missingness (37.3%)	0.373	PMM MI	2 vs 0	70.268	13.148	39.404	101.133	61.729
Natural missingness (37.3%)	0.373	PMM MI	3 vs 0	52.243	12.866	22.012	82.474	60.462
Natural missingness (37.3%)	0.373	PMM MI	3 vs 0	52.243	12.867	22.038	82.448	60.410
Natural missingness (37.3%)	0.373	Propensity−score MI	1 vs 0	54.395	12.510	24.817	83.972	59.155
Natural missingness (37.3%)	0.373	Propensity−score MI	1 vs 0	54.395	12.615	24.595	84.195	59.600
Natural missingness (37.3%)	0.373	Propensity−score MI	2 vs 0	66.215	13.370	34.606	97.824	63.218
Natural missingness (37.3%)	0.373	Propensity−score MI	2 vs 0	66.215	13.450	34.443	97.987	63.544
Natural missingness (37.3%)	0.373	Propensity−score MI	3 vs 0	45.437	11.760	17.635	73.239	55.605
Natural missingness (37.3%)	0.373	Propensity−score MI	3 vs 0	45.437	11.792	17.582	73.292	55.710
Natural missingness (37.3%)	0.373	Regression MI	1 vs 0	63.747	12.989	33.187	94.307	61.120
Natural missingness (37.3%)	0.373	Regression MI	1 vs 0	63.747	13.082	33.001	94.493	61.492

ACTG175_natural_missing_results.csv

scenario	missing_rate	method	contrast	estimate	se	lower	upper	width
Natural missingness (37.3%)	0.373	Regression MI	2 vs 0	71.007	12.770	40.963	101.051	60.088
Natural missingness (37.3%)	0.373	Regression MI	2 vs 0	71.007	12.887	40.718	101.296	60.577
Natural missingness (37.3%)	0.373	Regression MI	3 vs 0	49.797	12.869	19.521	80.073	60.552
Natural missingness (37.3%)	0.373	Regression MI	3 vs 0	49.797	12.893	19.493	80.101	60.608

ACTG175_natural_missing_results.csv

significant	shrink_alpha	variance_method
TRUE	NA	Non−shrinkage
TRUE	1.000	Shrinkage
TRUE	NA	Non−shrinkage
TRUE	1.000	Shrinkage
TRUE	NA	Non−shrinkage
TRUE	1.000	Shrinkage
TRUE	NA	Non−shrinkage
TRUE	0.730	Shrinkage
TRUE	NA	Non−shrinkage
TRUE	0.730	Shrinkage
TRUE	NA	Non−shrinkage
TRUE	0.730	Shrinkage
TRUE	NA	Non−shrinkage
TRUE	0.811	Shrinkage
TRUE	NA	Non−shrinkage
TRUE	0.811	Shrinkage
TRUE	NA	Non−shrinkage
TRUE	0.811	Shrinkage
TRUE	NA	Non−shrinkage
TRUE	0.726	Shrinkage

ACTG175_natural_missing_results.csv

significant	shrink_alpha	variance_method
TRUE	NA	Non−shrinkage
TRUE	0.726	Shrinkage
TRUE	NA	Non−shrinkage
TRUE	0.726	Shrinkage

ACTG175_shrinkage_width_gain.csv

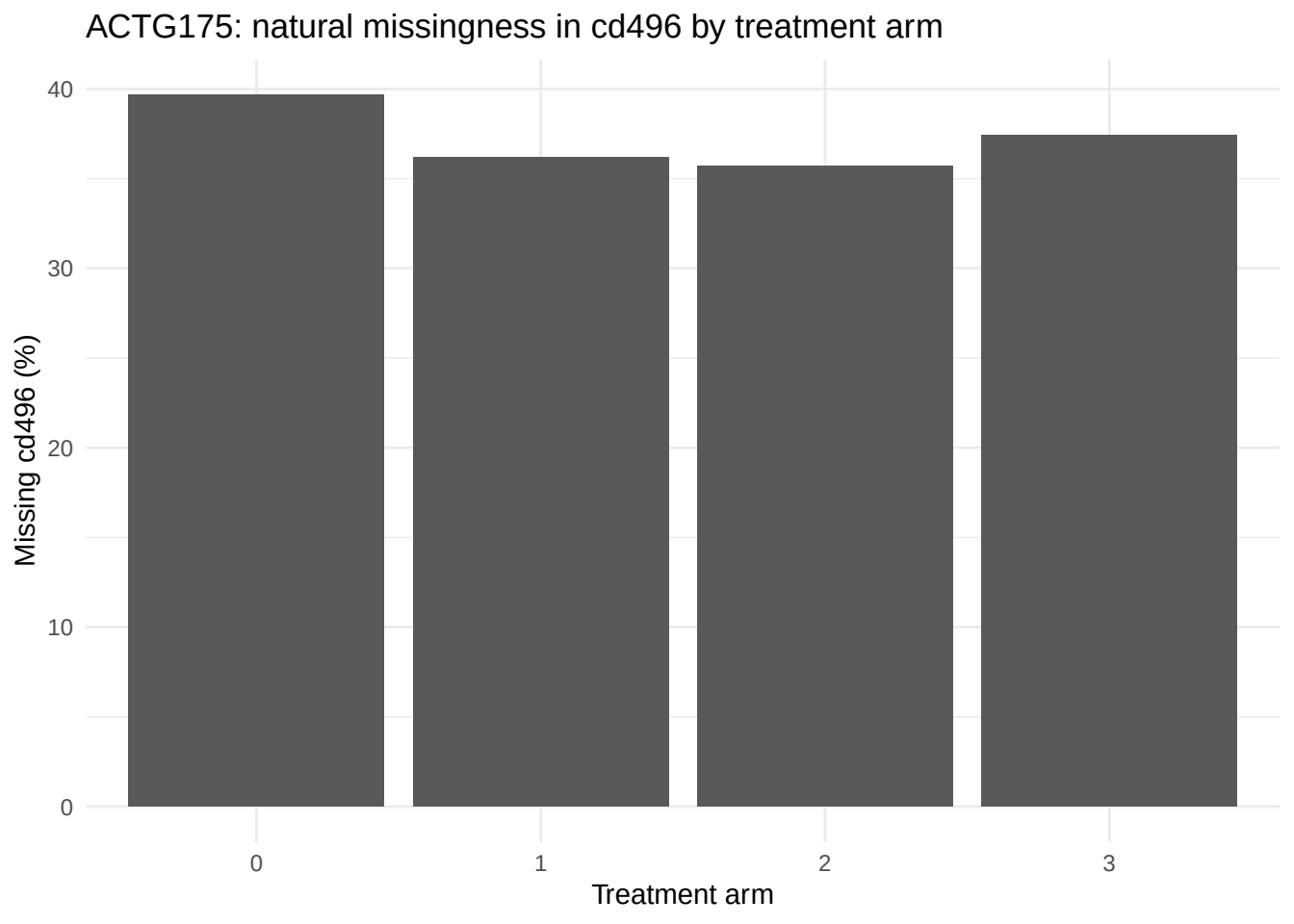
method	contrast	width_nonshrinkage	width_shrinkage	width_reduction_pct
PMM MI	1 vs 0	61.345	61.639	−0.479
PMM MI	2 vs 0	61.186	61.729	−0.888
PMM MI	3 vs 0	60.462	60.410	0.086
Propensity−score MI	1 vs 0	59.155	59.600	−0.753
Propensity−score MI	2 vs 0	63.218	63.544	−0.515
Propensity−score MI	3 vs 0	55.605	55.710	−0.190
Regression MI	1 vs 0	61.120	61.492	−0.608
Regression MI	2 vs 0	60.088	60.577	−0.813
Regression MI	3 vs 0	60.552	60.608	−0.092

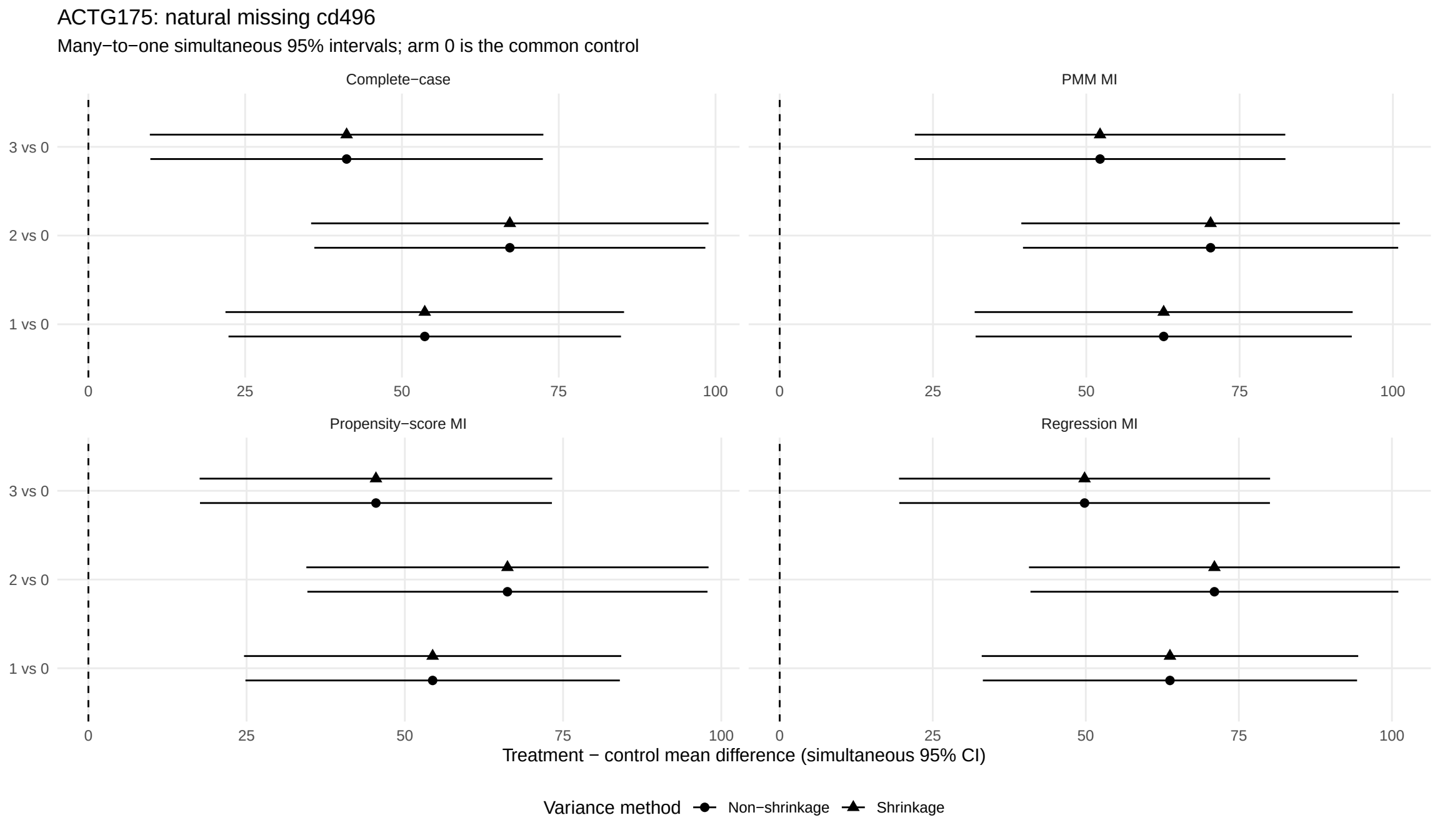
ACTG175_validation_benchmark_contrasts.csv

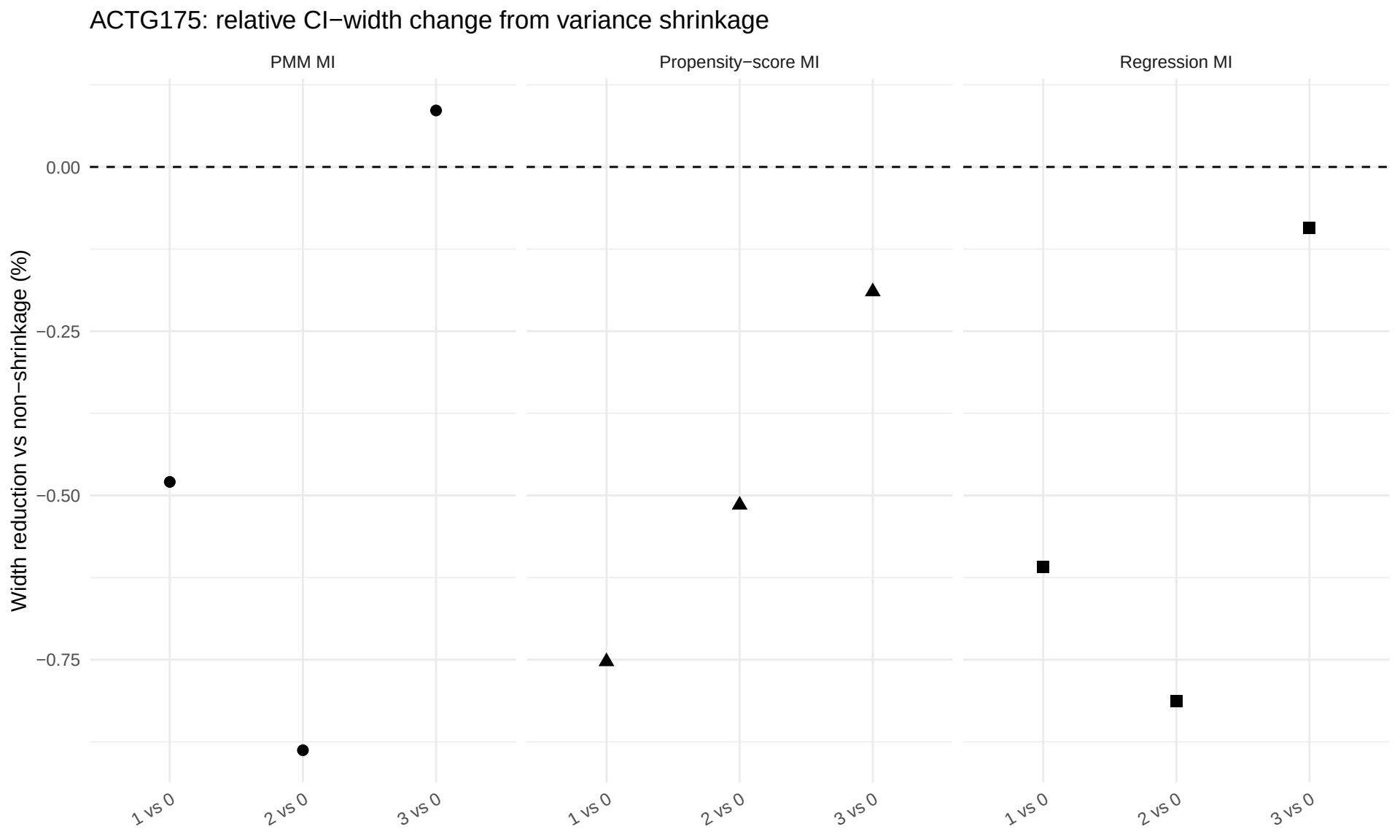
contrast	validation_complete_contrast
1 vs 0	53.635
2 vs 0	67.202
3 vs 0	41.175

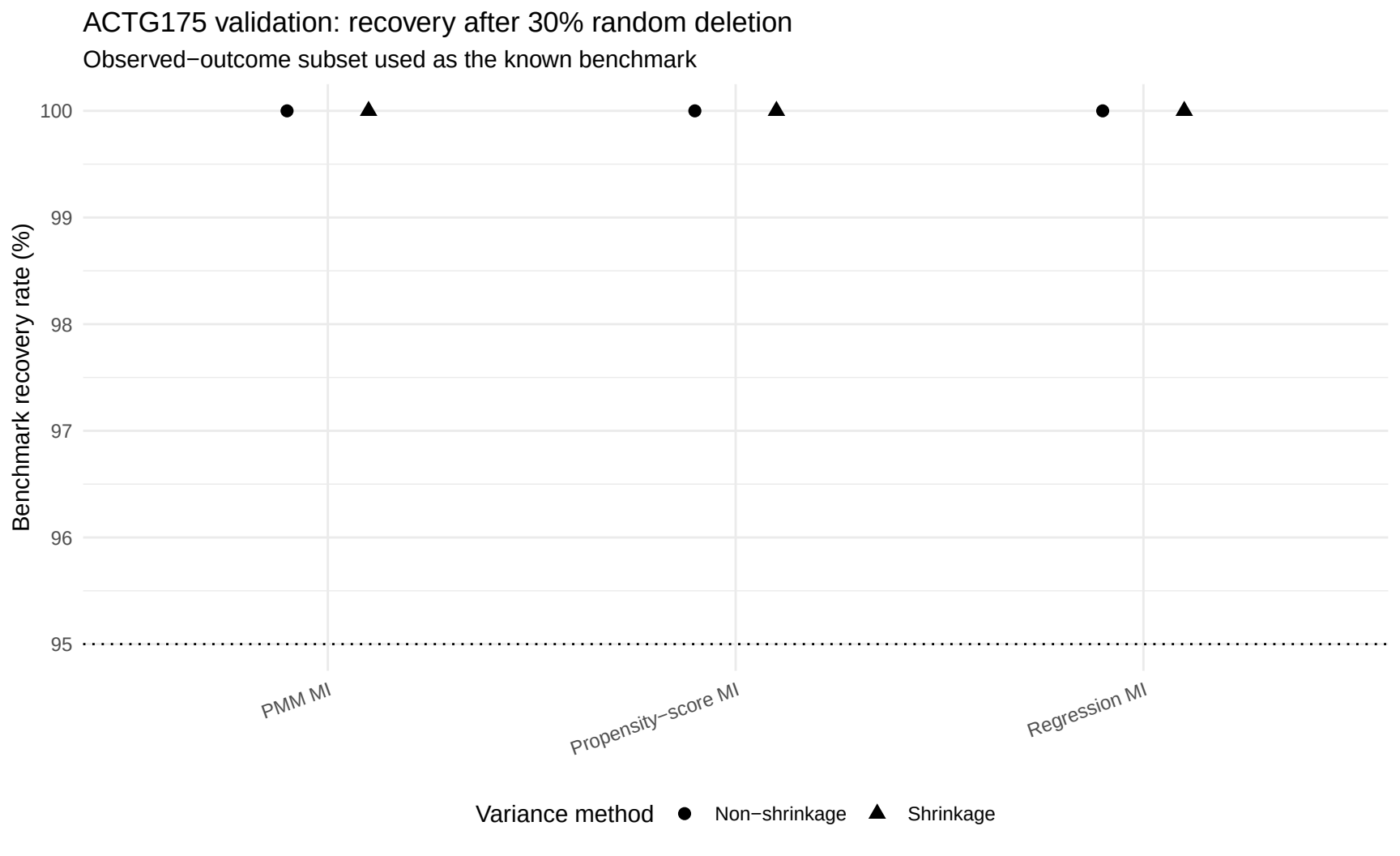
ACTG175_validation_summary.csv

missing_rate	method	variance_method	n_rows	rmse	mean_abs_error	benchmark_recovery_pct	avg_ci_width	mean_shrink_alpha
0.3	PMM MI	Non−shrinkage	3000	7.365	5.824	100	71.456	NA
0.3	PMM MI	Shrinkage	3000	7.365	5.824	100	71.850	0.729
0.3	Propensity−score MI	Non−shrinkage	3000	8.501	6.753	100	74.058	NA
0.3	Propensity−score MI	Shrinkage	3000	8.501	6.753	100	74.416	0.714
0.3	Regression MI	Non−shrinkage	3000	7.216	5.707	100	71.887	NA
0.3	Regression MI	Shrinkage	3000	7.216	5.707	100	72.280	0.736


ACTG175: natural missingness in cd496 by treatment arm
Missing cd496 (%)
40
30
20
10
0
0
1
2
3
Treatment arm


ACTG175: natural missing cd496
Many−to−one simultaneous 95% intervals; arm 0 is the common control
Complete−case
PMM MI
Propensity−score MI
Regression MI
3 vs 0
2 vs 0
1 vs 0
0
25
50
75
100
Treatment − control mean difference (simultaneous 95% CI)
Variance method
Non−shrinkage
Shrinkage


ACTG175: relative CI−width change from variance shrinkage
PMM MI
Propensity−score MI
Regression MI
Width reduction vs non−shrinkage (%)
0.00
−0.25
−0.50
−0.75
1 vs 0
2 vs 0
3 vs 0


ACTG175 validation: recovery after 30% random deletion
Observed−outcome subset used as the known benchmark
Benchmark recovery rate (%)
100
99
98
97
96
95
PMM MI
Propensity−score MI
Regression MI
Variance method
Non−shrinkage
Shrinkage